Muhammad Saad

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Skills

Programming Languages: C++ |Python |Scikit-Learn | NumPy |Pandas |Matplotlib
Domain Expertise: SolidWorks | ANSYS | MATLAB | Machine Learning| Data Analysis | COMSOL
Mathematics Foundations: Calculus, Probability & Statistics, Linear Algebra, Numerical Computing
Tools & Platforms: 3DExperiences | Git | GitHub | Latex |MS Office (MS Word, MS Outlook, MS PowerPoint, MS Teams)
Soft Skills: Documentation |Communication | Teamwork | Problem Solving | Willingness to Learn | Time management

Professional Experience

Mechanical Intern | Atlas Honda Limited, Sheikhupura, Pakistan **06/2026 – 07/2026**
Project Based Internship | SolidWorks & ANSYS
Feasibility Analysis and Design of Assembled Engine Lifter
Objective: Applied reverse engineering techniques to optimize the engine lifter design respect to the checkpoints.

- **Observed** engine assembly operations, including torquing, fitting, and inspection processes, to understand production flow and quality control practices.
- **Optimized** the engine lifter through **CAD redesign and FEA**, achieving a **15% weight reduction** while maintaining structural performance.
- **Enhanced** the lifter mechanism's structural safety through **FEA-driven redesign**, achieving a **1.5× improvement in factor of safety** under operating loads
- **Improved** operator ergonomics by redesigning the lifter operation for safer handling, reducing **REBA** score from **7** to **3** and lowering ergonomic risk during engine handling.

Mechanical Intern | Malakand-III Hydropower Plant, Pakistan **06/2025 – 07/2025**

Objective: To gain practical understanding of **hydropower generation, turbine operations, and plant maintenance** through hands-on exposure to powerhouse systems and equipment.

- **Gained hands-on exposure** to hydropower plant operations, working alongside engineers and technicians to understand power generation, plant equipment, and operational practices.
- **Analyzed** the operation of **turbines, generators, compressors, and auxiliary systems**, assessing their integration within the hydropower generation process.
- **Examined** turbine and generator **maintenance, inspection, lubrication, and cooling systems**, gaining practical insight into equipment reliability and preventive maintenance.
- **Evaluated** plant operating parameters and equipment performance to understand **efficiency, fault detection, and safe operation** of hydropower systems.

Education

Bachelor of Science in Mechanical Engineering — Ghulam Ishaq Khan Institute of Engineering Sciences and Technology — *Topi, Swabi, Pakistan* — **09/2023 - 06/2027**
CGPA [3.25/4]

- **Relevant Coursework:** Applied Physics, Computing and AI, Object Oriented Programming and Design, Thermodynamics-1, Thermodynamics-2, Statics, Dynamics, Mechanics of Solids-1, Mechanics of Solids-2, Fluid Mechanics-1, Fluid Mechanics-2, Design of Machine Element – I, Design of machine Elements-2, Mechanical Vibration, Heat Transfer, Numerical Analysis, Manufacturing Processes

Selected Projects

Explainable Deep Learning Framework for Multiclass and Multilabel Land Use and Land Cover Recognition using Remote Sensing Images
Undergraduate Thesis Project (FYP) | Aug 2026 – March 2027 (Ongoing)

- **Developed** an AI-driven non-destructive inspection system to detect **polyurethane foam insulation anomalies** in refrigerator cabinets during manufacturing.
- **Established** a baseline dataset by integrating **pressure, thermal, and vibration sensors** on test rigs to monitor normal refrigeration operation.
- **Extracted** thermodynamic features including **cream time, gel time, and tank temperatures** to identify early-stage insulation-related thermal anomalies.
- **Evaluated** neural network architectures for anomaly classification and integrated the optimal model into a **real-time diagnostic dashboard** to improve detection and reduce material waste.

Two-Stage Helical Gearbox with Bearing Fault Diagnosis
Machine Design Project | April 2026

- **Designed** and fabricated a two-stage helical gearbox using **10 mm shafts, 6000-series bearings**, and **3D-printed gears** for a **12:1** total speed ratio.

- **Integrated** accelerometers and sound sensors to capture vibration data from **healthy and intentionally faulted** bearing conditions.
- **Extracted** time-domain features including **RMS and Kurtosis** to characterize bearing fault signatures.
- **Developed** a Random Forest diagnostic model achieving approximately **97% classification accuracy** for bearing defects.

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Heat Transfer Enhancement of Pot-Core Inductors through Geometric Optimization

Academic Project | March 2026 – May 2026

- **Investigated** thermal limitations in enclosed pot-core inductors using a **2D axisymmetric numerical model** to optimize forced convective cooling.
- **Developed** a conjugate heat transfer model in **COMSOL Multiphysics** to simulate coupled solid-fluid heat transfer and laminar flow.
- **Compared** baseline, externally finned, and expanded internal-gap configurations to evaluate their effects on **heat removal and flow behaviour**.
- **Optimized** the internal flow gap by **2.3 mm**, reducing maximum temperature by **11.0%** by improving forced-convection heat dissipation.

Investigation of Capillary-Driven Flow and Diffusion of Colors in Paper Microfluidics

Academic Project | December 2025

- **Investigated** capillary-driven flow and mixing behaviour by varying **channel orientation and Y-junction angles** in paper microfluidic strips.
- **Fabricated** four microfluidic configurations and introduced blue and yellow dyes to measure **flow-front positions and mixing points** over time.
- **Calculated** flow velocities and **Peclet numbers** using the **Lucas–Washburn law** to compare advection and diffusion across configurations.
- **Demonstrated** faster horizontal flow of **0.253 mm/s** compared with **0.0876 mm/s** vertically, while larger junction angles improved fluid mixing.

Design of Combined Rankine Cycle Power Plant for Heat Recovery

Thermodynamics Project | March 2025

- **Designed** a 150-kW combined gas turbine and vapor power plant to recover waste heat and reduce fuel consumption.
- **Selected** methane for the gas cycle and **HFO-1336mzz(Z)** for the vapor cycle based on thermal performance and environmental considerations.
- **Analyzed** component-level thermodynamic performance and **exergy destruction** using **Engineering Equation Solver (EES)** to identify major sources of system inefficiency.
- **Optimized** boiler pressure at **600–650 kPa**, achieving a theoretical **54% efficiency peak** while reducing fuel consumption by approximately **35%** compared with a standalone turbine.

Awards

- **Chief Minister Scholarship of Khyber Pakhtunkhwa** | *University Level* fully funded
- **Talent Scholarship** | *Quaid-e-Azam Group of Schools and Colleges (Pre-engineering) fully funded*

Certificates

- Certified SOLIDWORKS Design Professional **(Dassault Systèmes)**
- Certified SOLIDWORKS Additive Manufacturing Associate **(Dassault Systèmes)**
- Certified SOLIDWORKS Simulation Associate **(Dassault Systèmes)**
- Certified SOLIDWORKS xMold Associate **(Dassault Systèmes)**
- Certified SOLIDWORKS xDesign Associate **(Dassault Systèmes)**
- Applied Computational Fluid Dynamics Course **(Coursera)**
- Data Analytics Essentials **(Coursera)**

Volunteer & Leadership

Technical Lead | Team Techno (GIKI, Pakistan) ***10/2023–Present***

- Lead **robotic system design, manufacturing, and mechanical integration**, coordinating technical execution across the team.

Executive Member | Team Infinity (GIKI, Pakistan) ***11/2024–Present***

- Contribute to **Formula Student vehicle design, material selection, manufacturing, and safety compliance**.

Event Coordinator | Society of Mechanical Engineers of Pakistan (GIKI Chapter) ***10/2025–Present***

- **Organize technical events and workshops**, coordinating with students, faculty, and industry professionals.

Language Skills

- English – C1 (Advanced)
- Urdu – Native
- Pashto – Native